

Spatial Mass distributions and Galactic Stability

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The Problem

Via newtonian mechanics and a gaussian integral we expect the velocity distribution of galaxies to look like:
The data shows it follows a curve like:

 center

Whats been Tried

- Neither General relativity nor Newtonian physics explain the motion of galaxies

$$G = -\frac{GMm}{r^2}\hat{r}$$

$$R^\lambda_{\gamma\alpha}$$

sfljngdlfjljfdnvldjnvldjnfvdlnvldjnfvdlnflvdjnfljvndflj

What I am trying

Why I am trying it d

- light traveling through plasma

$$\omega^2 = \omega_p^2 + k^2 \iff E^2 = m^2 + p^2$$

- plasma frequency

$$\omega_p^2 = \frac{e^2 n}{\epsilon_0 m}$$

- n

Why I am trying it cont.

Light incident on plasma

where I'm at

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$$

(Gauss's law)

$$\nabla \cdot \mathbf{B} = 0$$

(No magnetic monopoles)

$$\nabla \times \mathbf{E} = - \frac{\partial \mathbf{B}}{\partial t}$$

(Faraday-Lenz law)

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

(Ampere-Maxwell Law)

animation

center

code

```
import pygame  
import numpy
```

Next steps

Ideas for future

Gravitational Lensing

Appendix

Bibliography, References, Appendix. etc.

References

- 1.
7. HEIDEGGER, Martin. **Sein und Zeit**. 11. ed. Tübingen: Max Niemeyer Verlag, 1967.



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beginning